॥ सा विद्या या विमुक्तये ॥
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CS101C: Introduction to Programming (Using C)

Instructors: Prof. Aikta Arya and Prof. Konjengbam Anand

CSE Department

Indian Institute of Technology, Dharwad

Week 6

Today's Class (7/9/2026)

A Strange Thing About Arrays

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We already know how to get the address of the first element:

```
int arr[5] = {10,20,30,40,50};
&arr[0]
```

But C has a little surprise...

A Strange Thing About Arrays

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We already know how to get the address of the first element:

```
int arr[5] = {10,20,30,40,50};
&arr[0]
```

But C has a little surprise...

Try this!

```
int *p = arr;
```

In most expressions:

`arr` $\longrightarrow$ `&arr[0]`

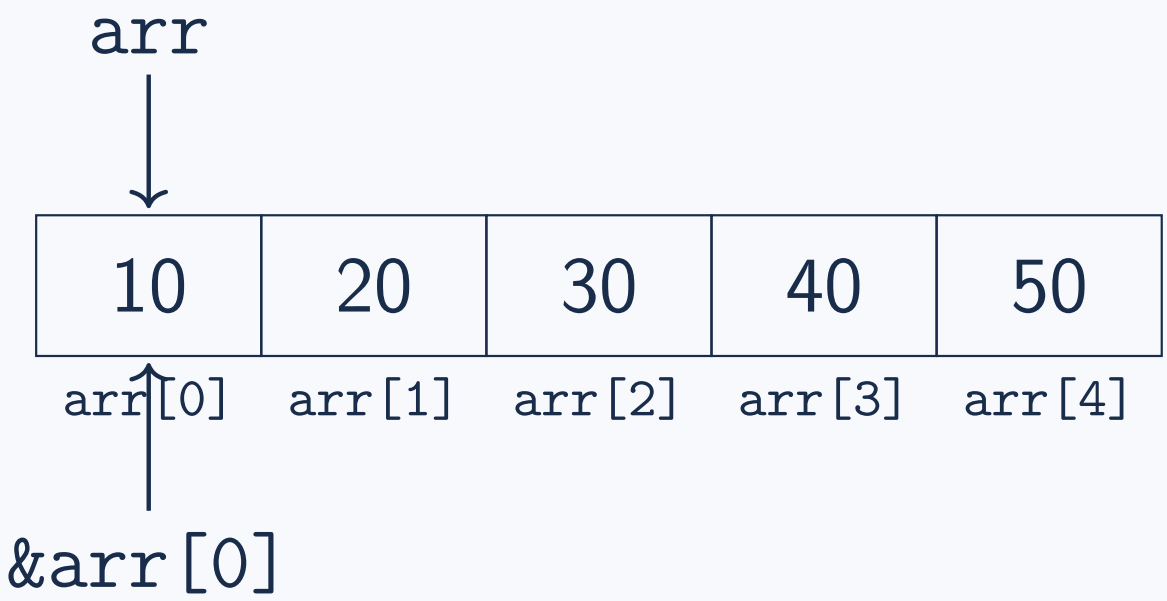
So these two are equivalent:

```
int *p = arr;
```

```
int *p = &arr[0];
```

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What does arr actually give us here?



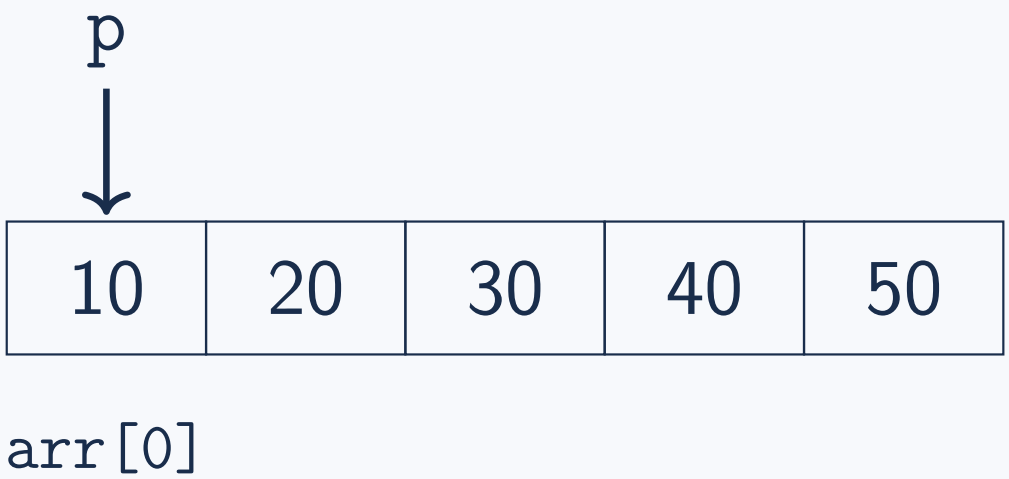
Array name is a synonym for the location of the first element.

`arr == &arr[0]`

Therefore: `int *p = arr;`

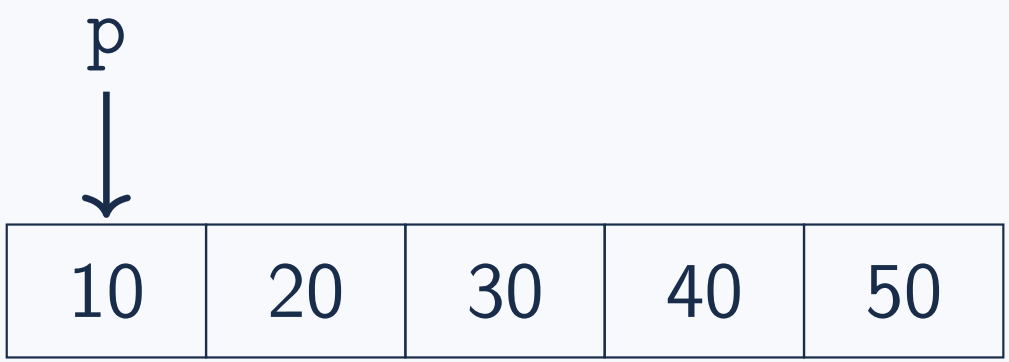
means `p` $\longrightarrow$ first element

So What Does *p Give?



So What Does *p Give?

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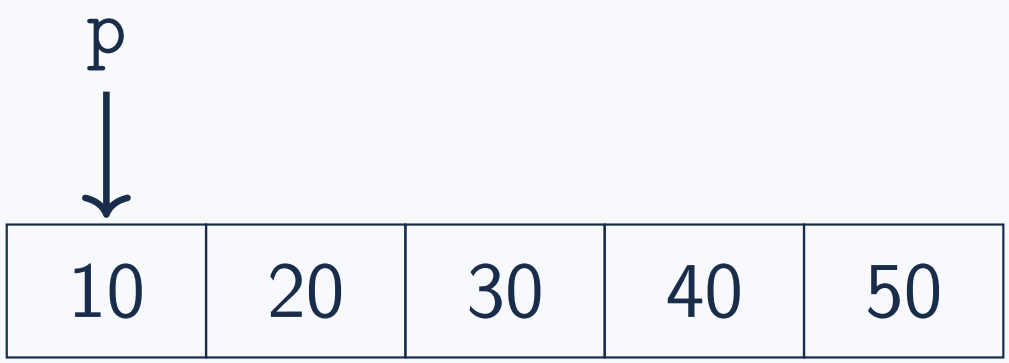


`arr[0]`

Since:

`p = arr = &arr[0]`

So What Does *p Give?



Since:

`p = arr = &arr[0]`

Dereferencing `p`:

`*p = arr[0]`

Remember

`p` = address `*p` = value

The Beautiful Relationship


```
arr[0] = *arr
```

The Beautiful Relationship

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```
arr[0] = *arr
```

Move one position:

```
arr[1] = *(arr + 1)
```

The Beautiful Relationship

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```
arr[0] = *arr
```

Move one position:

```
arr[1] = *(arr + 1)
```

```
arr[2] = *(arr + 2)
```

The Beautiful Relationship

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```
arr[0] = *arr
```

Move one position:

```
arr[1] = *(arr + 1)
```

```
arr[2] = *(arr + 2)
```

```
arr[3] = *(arr + 3)
```

The Beautiful Relationship

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```
arr[0] = *arr
```

Move one position:

```
arr[1] = *(arr + 1)
```

```
arr[2] = *(arr + 2)
```

```
arr[3] = *(arr + 3)
```

The Big Idea

Array indexing is closely connected to pointer arithmetic!

Let's Decode One Expression

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`arr[2]`

What if we rewrite it?

Let's Decode One Expression

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`arr[2]`

What if we rewrite it?

`arr[2]`

$\Downarrow$

`*(arr + 2)`

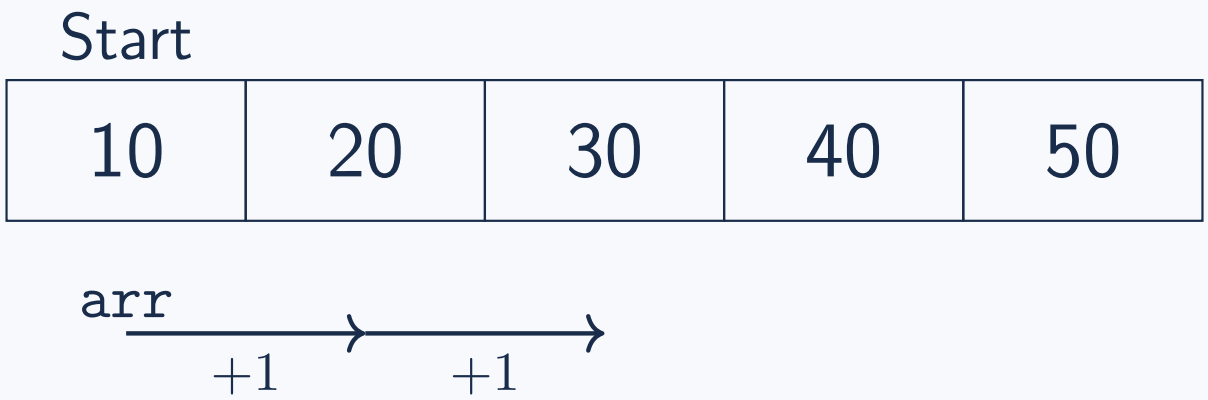
Think of it as:

$$\underbrace{\texttt{arr}}_{\text{start}} + \underbrace{2}_{\text{move 2 elements}} \longrightarrow *$$

Move first $\longrightarrow$ Dereference

Think of an Array as a Road

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Think of an Array as a Road

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Start

10	20	30	40	50



`arr + 0` $\rightarrow$ first element

Think of an Array as a Road

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Start

10	20	30	40	50

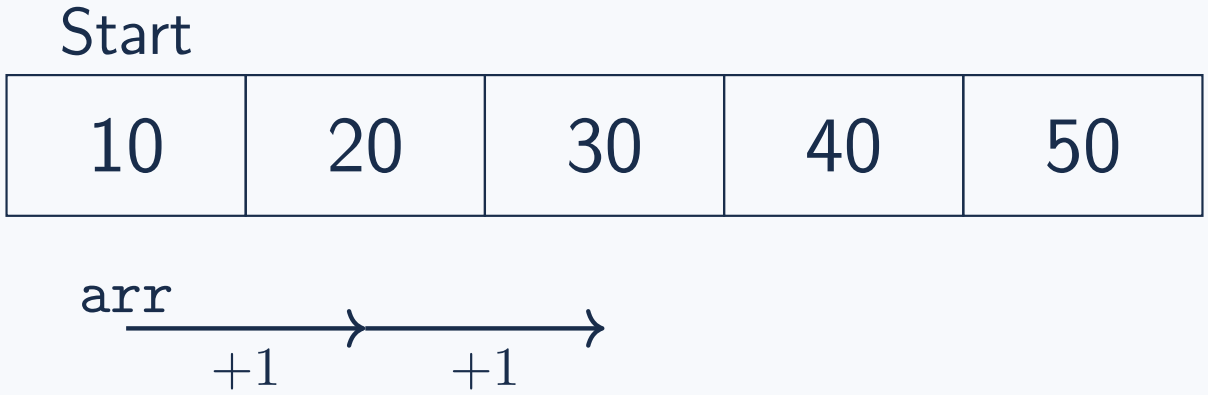


`arr + 0` $\rightarrow$ first element

`arr + 1` $\rightarrow$ next element

Think of an Array as a Road

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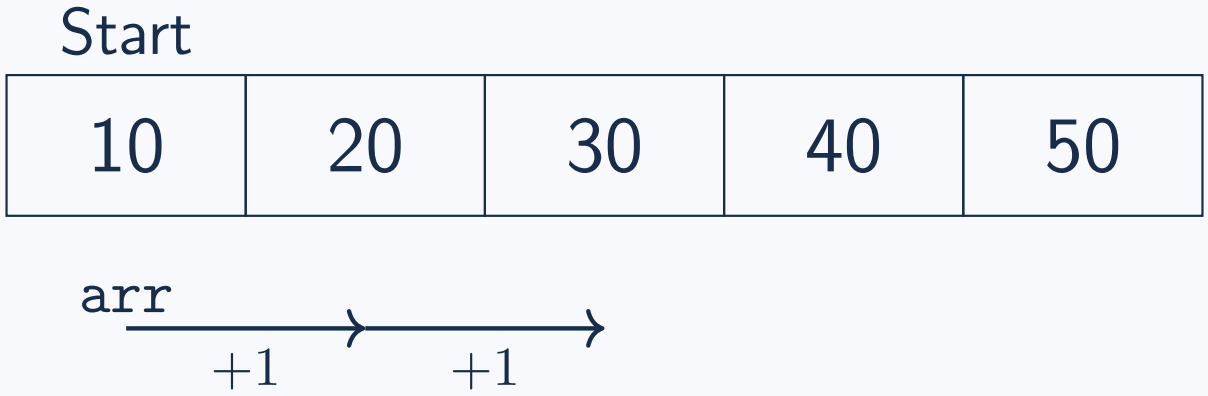


`arr + 0` $\rightarrow$ first element

`arr + 1` $\rightarrow$ next element

`arr + 2` $\rightarrow$ two elements forward

Think of an Array as a Road



`arr + 0` $\rightarrow$ first element

`arr + 1` $\rightarrow$ next element

`arr + 2` $\rightarrow$ two elements forward

Important

Pointer arithmetic moves by **elements**, not by arbitrary bytes.

Example!

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Given:

```
int arr[5] = {10,20,30,40,50};

int *p = arr;
```

What will each expression produce?

`*p`

`*(p+1)`

`*(p+2)`

`*(p+3)`

Reveal!

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Expression	Value
`*p`	10
`*(p+1)`	20
`*(p+2)`	30
`*(p+3)`	40

Reveal!

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Expression	Value
`*p`	10
`*(p+1)`	20
`*(p+2)`	30
`*(p+3)`	40

One Pattern to Remember

```
arr[i] = *(arr+i)
```

Reveal!

॥ सा विद्या या विमुक्तये ॥
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Expression	Value
`*p`	10
`*(p+1)`	20
`*(p+2)`	30
`*(p+3)`	40

One Pattern to Remember

```
arr[i] = *(arr+i)
```

This is the beautiful connection between **arrays** and **pointers**.

Is This a String?

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```
char arr[5] = {'H','E','L','L','O'};
```

'H'	'E'	'L'	'L'	'O'

Question:
Is `arr` the string "HELLO"?

Think!

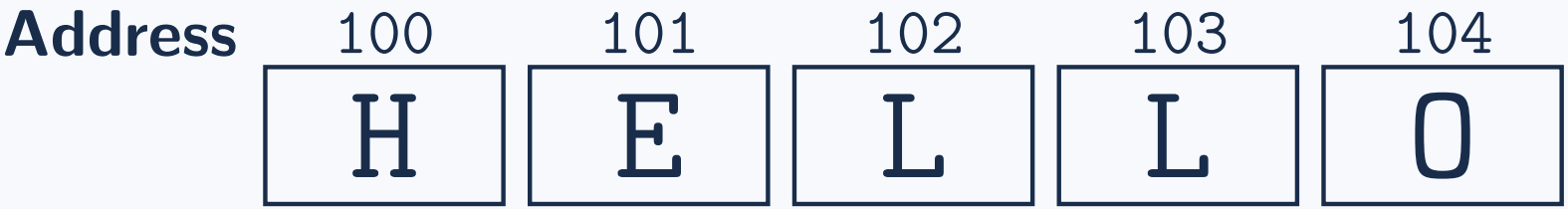
YES or NO?

Let's investigate...

Look Inside Memory

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What is actually stored in memory?



Think!

Where is the '\0'?

There isn't one!

And THAT is the problem.

The Secret Ingredient

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CHARACTER ARRAY

'H'	'E'	'L'	'L'	'O'

VS

STRING

'H'	'E'	'L'	'L'	'O'	'\0'

A string is a sequence of characters that ends with a NULL CHARACTER

Memory

H	E	L	L	O	\0
100	101	102	103	104	105

Important Terminology

'\0' $\longrightarrow$ **NULL CHARACTER** $\longrightarrow$ **STRING TERMINATOR**

How Do We Declare a String?

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A string is declared using a character array.

Method 1 — Character by Character

```
char name[6] = {'H','E','L','L','O','\0'};
```

Method 2 — Using a String Literal

```
char name[6] = "HELLO";
```

Method 3 — Let C Count for You

```
char name[] = "HELLO";
```

Key Idea

`"HELLO"` $\longrightarrow$ `'H' 'E' 'L' 'L' 'O' '\0'`

What Does %s Actually Do?

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Does `printf` magically know the length of a string?

START $\longrightarrow$



The %s Robot

$$\texttt{H} \rightarrow \texttt{E} \rightarrow \texttt{L} \rightarrow \texttt{L} \rightarrow \texttt{O} \rightarrow \texttt{'\textbackslash 0'}$$

"Print the next character..."

"Print the next character..."

"Print the next character..."

"OH! I found '\0'!"

STOP!

`%s` keeps reading until it encounters '\0'.

The TRAP Question!!

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Predict the Output!

```
int main()
{
    char arr[5] = {'H','E','L','L','O'};

    char *ptrC = arr;

    for(int i = 0; i < 5; i++)
    {
        printf("%c", *ptrC);
        ptrC++;
    }

    printf("%s", ptrC);
}
```

Q1.

What will the `for` loop print?

The TRAP Question!!

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Predict the Output!

```
int main()
{
    char arr[5] = {'H','E','L','L','O'};

    char *ptrC = arr;

    for(int i = 0; i < 5; i++)
    {
        printf("%c", *ptrC);
        ptrC++;
    }

    printf("%s", ptrC);
}
```

Q1.

What will the `for` loop print?

HELLO

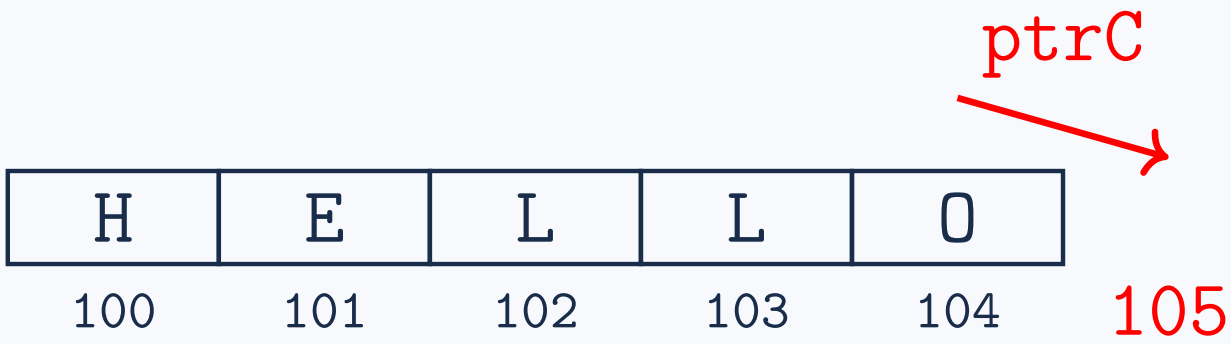
Q2.

After the loop, where is `ptrC` pointing?

THE TRAP!

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After the loop:



What happens now?

```
printf("%s", ptrC);
```

Where is the '\0'?

DANGER

`ptrC` does **NOT** point to a null-terminated string.

Therefore, `printf("%s", ptrC)` has **undefined behavior**.

It might print garbage, print unexpected characters, or even crash.

Another TRAP!

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What will this print?

The Mystery

```
int main()
{
    char arr[5] =
        {'H','E',300,'L','O'};

    char *ptrC = arr;

    printf("%s", ptrC);
}
```

Vote!

A) HELLO

B) HE?LO

C) HE + garbage

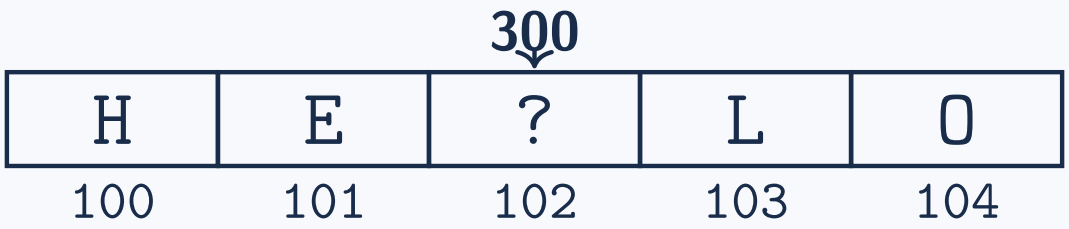
D) Undefined behavior

Don't run it yet!

Think about what is happening in memory.

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Why can't we predict the output?



Problem 1: No \0!

H E [300] L O

%s keeps reading until it finds a null character.

But there isn't one!

Problem 2: What is stored?

$300 \longrightarrow$ char

On a typical 8-bit char system, the value cannot be represented directly as the integer 300.

The exact result is **implementation-defined**.

```
printf("%s", ptrC);
```

Undefined Behavior

Safest answer: We cannot predict the output.

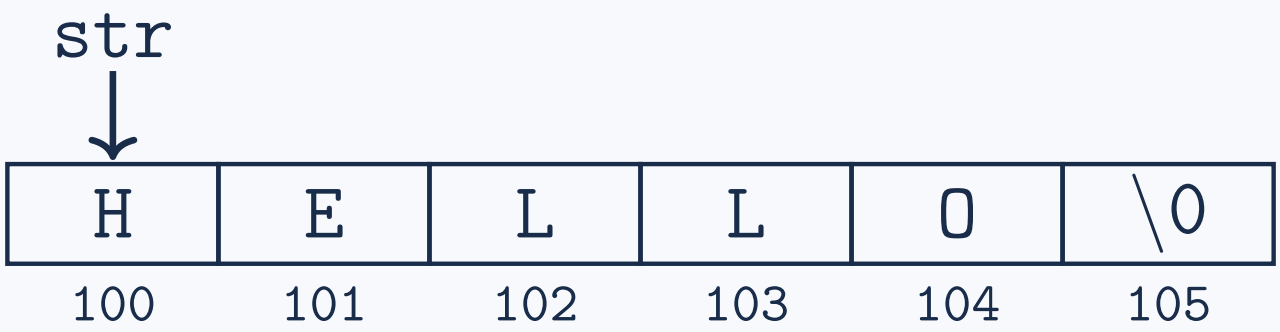
॥ सा विद्या या विमुक्तये ॥
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```
char *str = "HELLO";
```

What is `str`?

A POINTER!

And what does it point to?



`str` $\longrightarrow$ **address of 'H'**

Let's Dereference It

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```
char *str = "HELLO";
```

H	E	L	L	O	\0

What does this print?

```
printf("%c", *str);
```

⇓

H

Now move one position

```
printf("%c", *(str + 1));
```

⇓

E

`*str`	⟶ **H**
`*(str + 1)`	⟶ **E**
`*(str + 2)`	⟶ **L**
`*(str + 3)`	⟶ **L**
`*(str + 4)`	⟶ **O**

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Consider this:

```
char *str = "HELLO";
*str = 'Y';
```

What should happen?

HELLO

H	E	L	L	O

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Consider this:

```
char *str = "HELLO";
*str = 'Y';
```

What should happen?

HELLO

H	E	L	L	O

"HELLO becomes YELLO!"

The Reveal

Modifying a string literal has **undefined behavior**

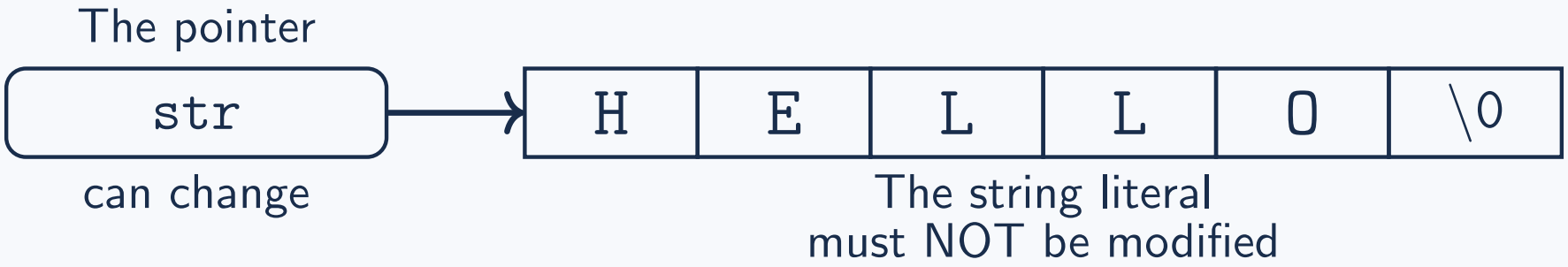
It may cause a segmentation fault, or behave in some other way.

One Important Distinction: Pointer vs. Data



Look Carefully

```
char *str = "HELLO";
```



Key Idea

The pointer is not the same thing as the characters it points to!

`str = "BYE";` ✓ Pointer can point somewhere else `*str = 'Y';` ×

Do not modify the string literal

String literals are not modifiable. If we want a string that we can modify, use a character array.

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When a string should not be modified:

```
const char *str = "HELLO";
```

`const` $\longrightarrow$ Do not modify

```
*str = 'Y';          //  Not allowed
```

The compiler can now catch this mistake.

Safe Rule to Remember

```
const char *str = "HELLO";
```

Use `const` when the pointer is meant to read, not modify, the string.

Entry Question: Can You Reverse an Array?

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Given

```
int arr[5] = {10, 20, 30, 40, 50};
```

How many different ways can you reverse it?

1. **Use a duplicate array**
   Recall: `arr2[10]`
2. **Use only ONE temporary variable**
   Recall: `tmp = arr[i]`
3. **Use NO temporary variable**
   Recall: `a=a+b; b=a-b; a=a-b;`
4. **Use Pointers**

Bonus: Can you do it using multiple expressions in a `for` loop?

```
for(i=0,j=4; i<5 && j>=0; i++,j--)
```

Today's Class (9 /9/2026)


Imagine This...
You are playing cards.
Someone gives you cards
one at a time.


1
First card:
7


Imagine This...
You are playing cards.
Someone gives you cards one at a time.
1
First card:
7
2
You already have: 7
New card arrives: 3
3


Imagine This...
You are playing cards.
Someone gives you cards one at a time.
1
First card:
7
2
You already have: 7
New card arrives: 3
3
3
3
Where would you put 3?


Imagine This...
You are playing cards.
Someone gives you cards one at a time.
1
First card:
7
2
You already have: 7
New card arrives: 3
3
3
Where would you put 3?
4
After thinking...
here's one possible place!
3
7
So...
3 7

Another Card Arrives





Another Card Arrives





Another Card Arrives





Another Card Arrives

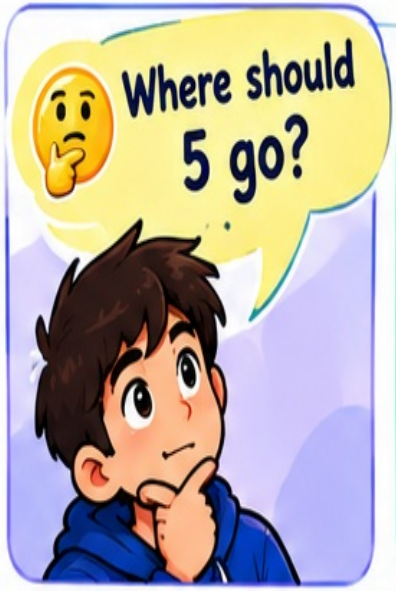


You have:

3 ♣ 7 ♥

New card = 5

5 ♥



Answer:

3 ♣ 5 ♥ 7 ♥



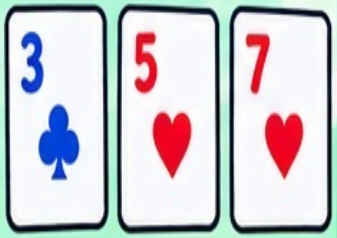
Did we rearrange the entire hand?

No!

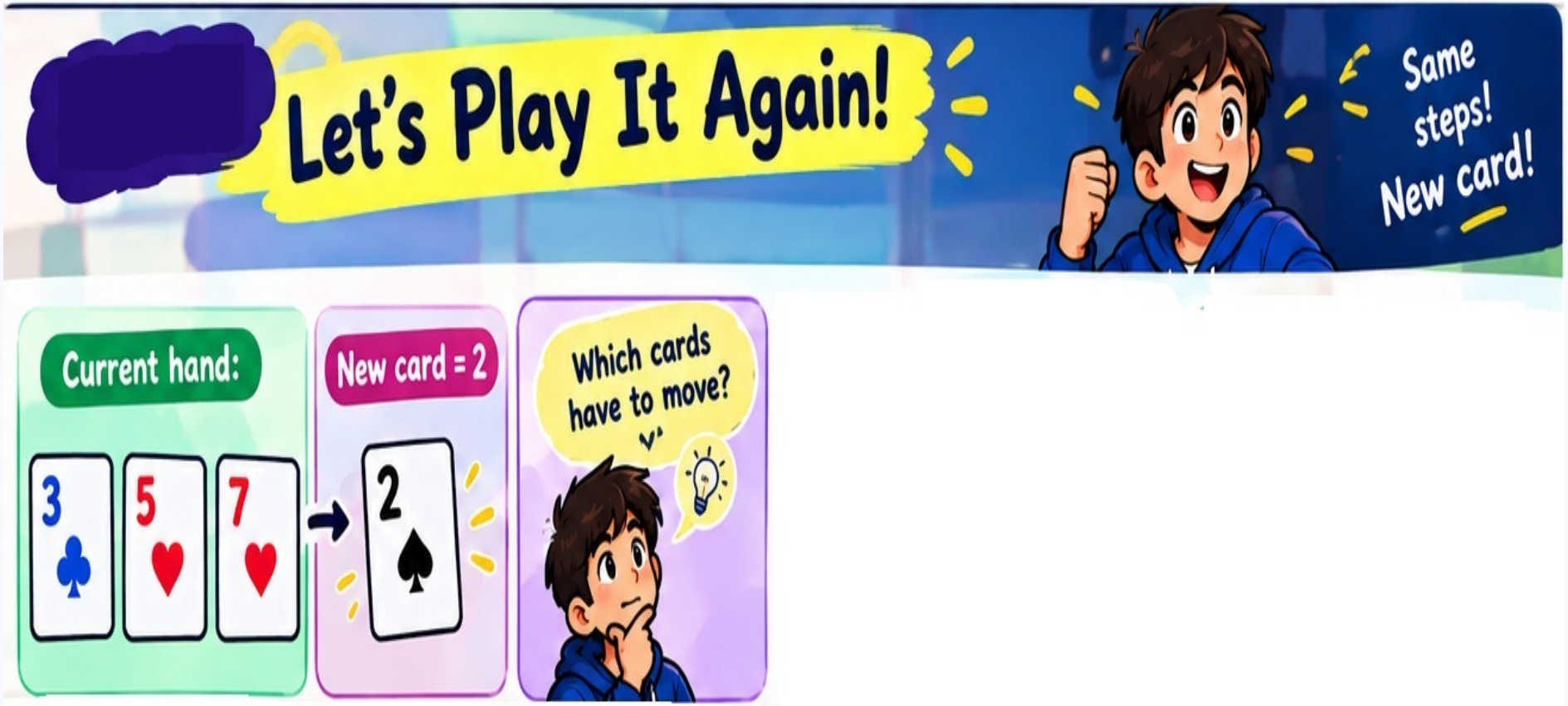
We only moved the cards that were **bigger than 5.**



Current hand:



Let's Play It Again!



Let's Play It Again!



Same steps! New card!



Let's Play It Again!



Same steps! New card!

Current hand:



New card = 2



Which cards have to move?



Move these to the right:



3 → right | 5 → right | 7 → right

New hand:

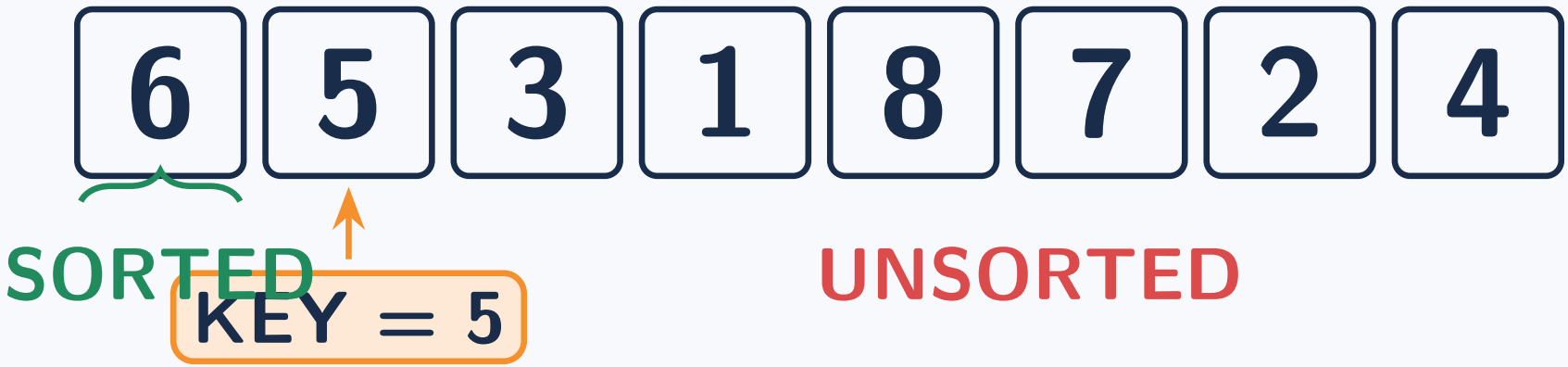


We take one new element and **INSERT** it into its correct position.

That's why it is called **INSERTION SORT.**

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY $\rightarrow$ take ONE element from the unsorted part
$\rightarrow$ insert it into the correct position

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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YES!

$5 < 6$?



KEY → **take ONE element from the unsorted part**
→ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY → take ONE element from the unsorted part
→ insert it into the correct position

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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INSERT 5 HERE!



KEY $\rightarrow$ **take ONE element from the unsorted part**
$\rightarrow$ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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YES!

$3 < 6$ **?**



KEY $\rightarrow$ **take ONE element from the unsorted part**
$\rightarrow$ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!



KEY $\rightarrow$ **take ONE element from the unsorted part**
$\rightarrow$ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!

3	5	6	1	8	7	2	4

KEY $\rightarrow$ **take ONE element from the unsorted part**
$\rightarrow$ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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sorted!

3 5 6

3	5	6	1	8	7	2	4

KEY $\rightarrow$ **take ONE element from the unsorted part**
$\rightarrow$ **insert it into the correct position**

Insertion Sort — Let's Watch It Happen!

$$\text{KEY} = 1$$



Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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$$\text{KEY} = 1$$



Insertion Sort — Let's Watch It Happen!

$$KEY = 1$$

3	1	5	6	8	7	2	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY = 1



Insertion Sort — Let's Watch It Happen!

$$\text{KEY} = 1$$

1	3	5	6	8	7	2	4

Insertion Sort — Let's Watch It Happen!

KEY = 8

1	3	5	6	8	7	2	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 8

1	3	5	6	8	7	2	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 7

1	3	5	6	8	7	2	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 7



Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY = 7

1	3	5	6	7	8	2	4

Insertion Sort — Let's Watch It Happen!

KEY = 2



Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY = 2



Insertion Sort — Let's Watch It Happen!

KEY = 2



Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY = 2



Insertion Sort — Let's Watch It Happen!

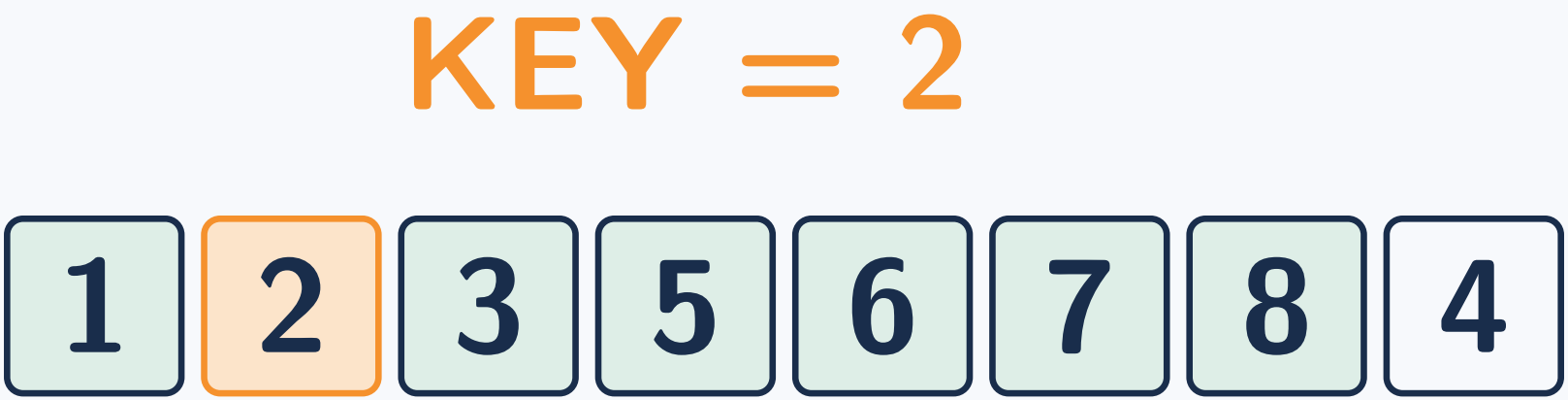
॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 2

1	3	2	5	6	7	8	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 2

1	2	3	5	6	7	8	4

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
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KEY = 4



Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 4

1	2	3	5	6	7	4	8

Insertion Sort — Let's Watch It Happen!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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KEY = 4

1	2	3	5	6	4	7	8

Insertion Sort — Let's Watch It Happen!

KEY = 4

1	2	3	5	4	6	7	8

Insertion Sort — Let's Watch It Happen!

KEY = 4

1	2	3	4	5	6	7	8

Insertion Sort — Let's Watch It Happen!

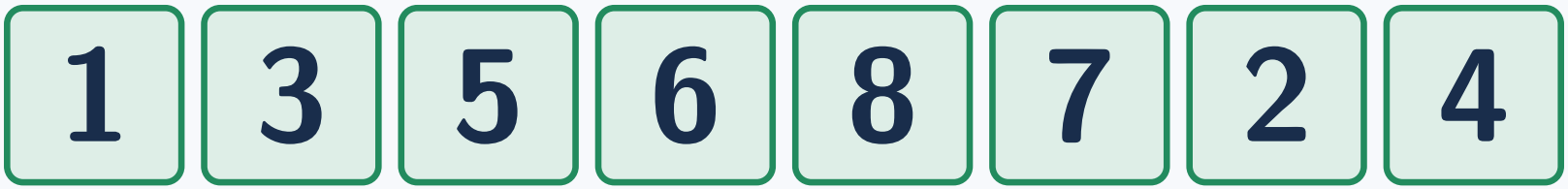
॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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$$\text{KEY} = 4$$

1	2	3	4	5	6	7	8

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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All elements are sorted!



Let's pause... What did we actually do?

Can you identify the operations?

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element
2. Store it somewhere

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element
2. Store it somewhere
3. Look at elements to its left

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element
2. Store it somewhere
3. Look at elements to its left
4. Move larger elements one position right

॥ सा विद्या या विमुक्तये ॥
संस्थान धारवाड
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Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element
2. Store it somewhere
3. Look at elements to its left
4. Move larger elements one position right
5. Put the element in the empty position

Let's pause... What did we actually do?

Can you identify the operations?

1. Pick an element
2. Store it somewhere
3. Look at elements to its left
4. Move larger elements one position right
5. Put the element in the empty position

PICK → STORE → COMPARE → SHIFT → INSERT

Complete C Program

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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```
#include <stdio.h>
int main()
{   int arr[] = {7, 3, 5, 2, 6};
    int n = 5;
    for (int i = 1; i < n; i++)
    {
        int key = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j] > key)
        {   arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Write a program that counts the total number of **shifts** performed by Insertion Sort.

For:

5	4	3	2	1

Your program should print:

```
Sorted array:  1 2 3 4 5

Total shifts:  ?
```

Today's Class (11/9/2026)
Practice session 1

POINTER TRAP #1

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30,40,50};

int *p = arr;

printf("%d", *(p + 3));
```

Don't run the program!

Q1.

What will be printed?

1. 20
2. 30
3. 40
4. 50

Your Vote!

A B C D

POINTER TRAP #1 — ANSWER

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Remember!

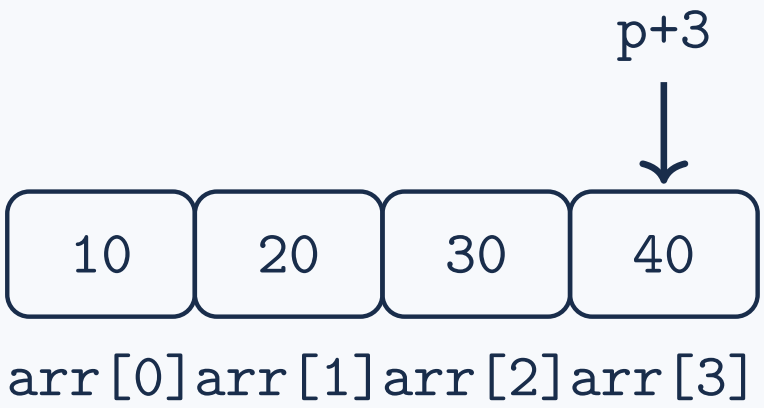
$$p = arr = \&arr[0]$$

Therefore,

$$p + 3 = \&arr[3]$$

and

$$*(p + 3) = arr[3]$$



$$\boxed{40}$$

Answer: C

POINTER TRAP 2

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = &arr[1];

printf("%d ", *(p+1));
printf("%d", *p+1);
```

Are these two expressions the same?

Q1.
What is the first value?

POINTER TRAP 2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = &arr[1];

printf("%d ", *(p+1));
printf("%d", *p+1);
```

Q1.
What is the first value?

30

Q2.
What is the second value?

Are these two expressions the same?

POINTER TRAP 2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = &arr[1];

printf("%d ", *(p+1));
printf("%d", *p+1);
```

Are these two expressions the same?

Q1.
What is the first value?

30

Q2.
What is the second value?

21

Why are they DIFFERENT?

॥ सा विद्या या विमुक्तये ॥
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Expression 1

$$\boxed{*(p+1)}$$

First move:

$$p \rightarrow p+1$$

Then dereference:

$$*(p+1) = 30$$

Expression 2

$$\boxed{*p+1}$$

First dereference:

$$*p = 20$$

Then add:

$$20+1 = 21$$

$$\boxed{*(p+1) \neq *p+1}$$

POINTER TRAP #3

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;

printf("%d ", *p++);
printf("%d", *p);
```

Where does `p` move?

Q1.
What does `*p++` mean?

1. `(*p)++`
2. `*(p++)`
3. `*(++p)`
4. `++(*p)`

POINTER TRAP #3

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;

printf("%d ", *p++);
printf("%d", *p);
```

Where does `p` move?

Q1.
What does `*p++` mean?

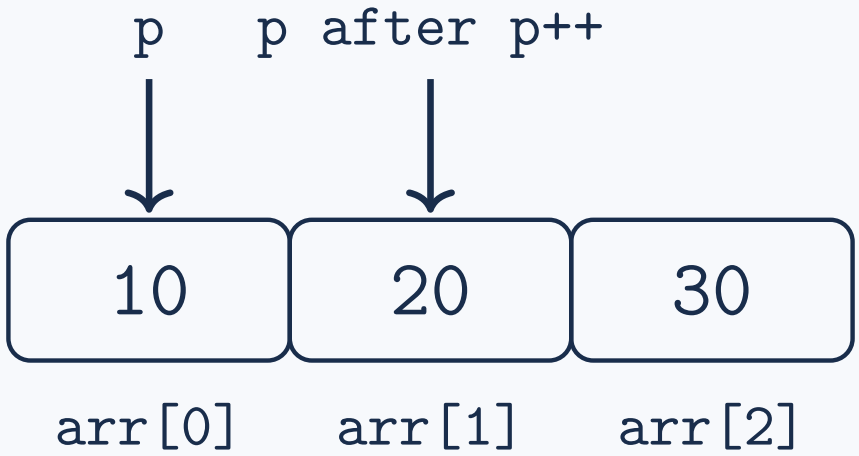
1. `(*p)++`
2. `*(p++)`
3. `*(++p)`
4. `++(*p)`

Answer

B

Let's TRACE the Pointer!

॥ सा विद्या या विमुक्तये ॥
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$$*p++ = *(p++)$$

Print 10	$\rightarrow$	move pointer	$\rightarrow$	$*p = 20$

$$10 \quad 20$$

Trap 3 — Decode the Four Options!

॥ सा विद्या या विमुक्तये ॥
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The same symbols... but completely different meanings!

1. `(*p)++`

VALUE changes

`p` stays here

`*p` increases by 1

`x = 10` ⇓

`(*p)++` ⇓

`x = 11`

2. `*(p++)`

POINTER moves

use old `p` first

then `p++`

`*p = 10` ⇓

print 10 ⇓

`p` moves to next element

Trap 3 — Decode the Four Options!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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The same symbols... but completely different meanings!

1. (*p)++

VALUE changes

p stays here
*p increases by 1
x = 10 ⇓
(*p)++ ⇓
x = 11

2. *(p++)

POINTER moves

use old p first
then p++
*p = 10 ⇓
print 10 ⇓
p moves to next element

3. *(++p)

MOVE FIRST

++p first
then *p
p → arr[0] ⇓
p moves to arr[1] ⇓
*p = 20

4. ++(*p)

VALUE changes FIRST

++ applied to *p
then use the new value *p =
10 ⇓
++(*p) ⇓
*p = 11

POINTER TRAP 4

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;

printf("%d ", ++*p);

printf("%d ", *p++);

printf("%d", *p);
```

Q1.
What does

```
++*p
```

mean?

POINTER TRAP 4

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;

printf("%d ", ++*p);

printf("%d ", *p++);

printf("%d", *p);
```

Q1.
What does

```
++*p
```

mean?

```
++(*p)
```

Q2.
What does

```
*p++
```

mean?

POINTER TRAP 4

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;

printf("%d ", ++*p);

printf("%d ", *p++);

printf("%d", *p);
```

Q1.
What does

```
++*p
```

mean?

`++(*p)`

Q2.
What does

```
*p++
```

mean?

`*(p++)`

POINTER TRAP 4 — SOLUTION

॥ सा विद्या या विमुक्तये ॥
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Initially:

$$arr = \{10, 20, 30\}$$

$$++*p = ++(*p)$$

Therefore:

$$10 \rightarrow 11$$

So first output:

$$\boxed{11}$$

POINTER TRAP 4 — SOLUTION

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Initially:

$$arr = \{10, 20, 30\}$$

$$++*p = ++(*p)$$

Therefore:

$$10 \rightarrow 11$$

So first output:

$$\boxed{11}$$

Now:

$$*p++ = *(p++)$$

Print current value: 11

Then move: $p : \ arr[0] \rightarrow arr[1]$

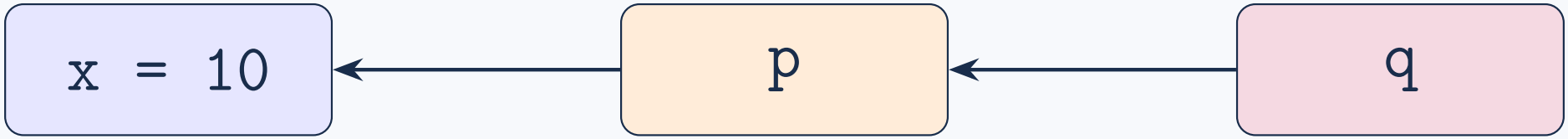
Finally: $*p = 20$

$$\boxed{11 \quad 11 \quad 20}$$

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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MULTIPLE POINTERS

Things are about to get confusing...



One variable...
Two pointers...
Then a pointer to a pointer!

Level 2 — Pointer Puzzle #1

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int x = 10;

int *p = &x;
int *q = &x;

*p = 20;

printf("%d %d %d", x, *p, *q);
```

VOTE!

What will be printed?

1. 10 20 10
2. 20 20 20
3. 10 10 10
4. 20 10 20

Level 2 — Pointer Puzzle #1

॥ सा विद्या या विमुक्तये ॥
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Predict the Output!

```
int x = 10;

int *p = &x;
int *q = &x;

*p = 20;

printf("%d %d %d", x, *p, *q);
```

VOTE!

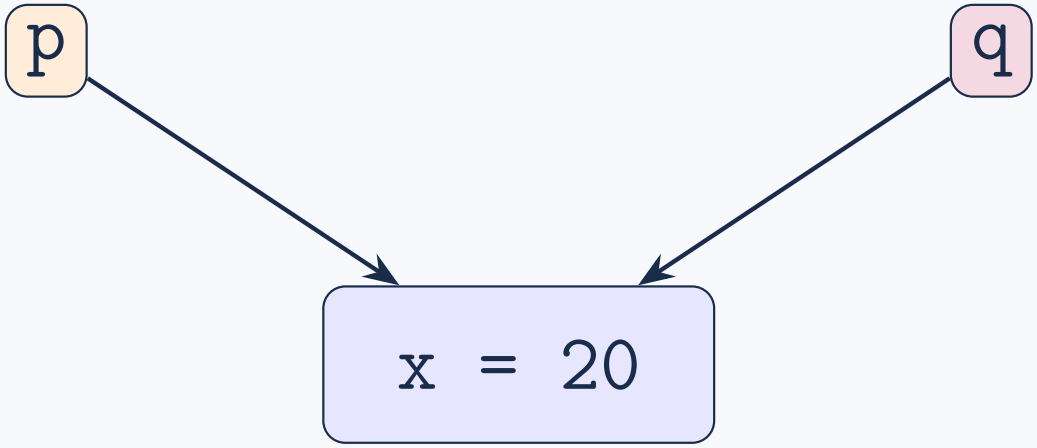
What will be printed?

1. 10 20 10
2. 20 20 20
3. 10 10 10
4. 20 10 20

Don't calculate!
Trace the arrows!

Solution!!

॥ सा विद्या या विमुक्तये ॥
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*p = 20 means:

x = 20

Since q also points to x:

*q = 20

20 20 20

Level 2 — Pointer Puzzle #2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int x = 10;

int *p = &x;
int **q = &p;

**q = 50;

printf("%d", x);
```

Q1.
What does q point to?

1. x
2. p
3. *p
4. 50

Level 2 — Pointer Puzzle #2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int x = 10;

int *p = &x;
int **q = &p;

**q = 50;

printf("%d", x);
```

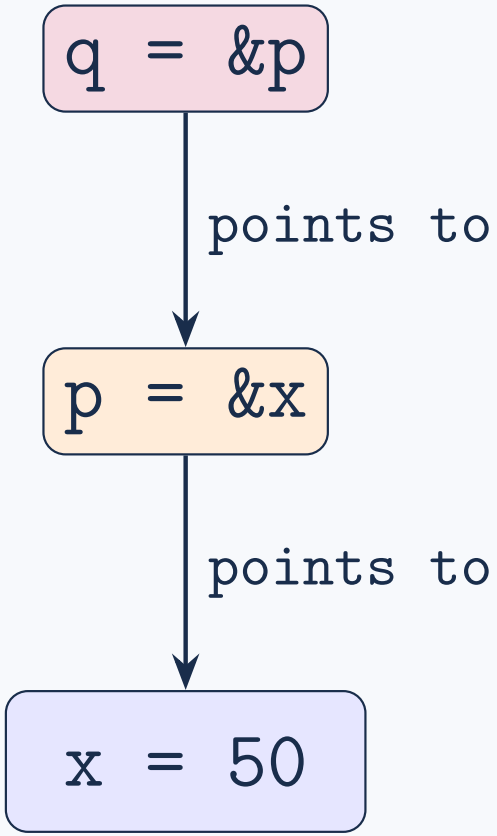
Q1.
What does q point to?

1. x
2. p
3. *p
4. 50

Q2.
What will x become?

1. 10
2. 50
3. Address of p
4. Undefined

Solution — The Double Dereference!



$$q \rightarrow p \rightarrow x$$

Therefore: $*q = p$ and $**q = x$

So: `**q = 50`

Output: 50

Level 2 — Pointer Puzzle #3

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int a = 10;
int b = 20;

int *p = &a;
int **q = &p;

*q = &b;

printf("%d", *p);
```

The Trap!

Initially:

$$p \rightarrow a$$

Then:

$$*q = \&b$$

What does p point to now?

1. a
2. b
3. q
4. Nothing

Solution — `**q` Can Change Where `p` Points!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Before:

$p \rightarrow a$

Then:

```
*q = &b
```

Remember:

$*q = p$

Therefore:

```
p = &b
```

Now:

$p \rightarrow b$

Therefore:

```
*p = 20
```

Output: 20

LEVEL 3

Now let's use your brain!

`*` + `++` + `p++` + `arrays`

POINTER Q #1

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30,40};

int *p = arr + 1;

*p = *p + 5;

printf("%d %d %d %d",
       arr[0], arr[1],
       arr[2], arr[3]);
```

Q1.
Where does p point?

1. arr[0]
2. arr[1]
3. arr[2]
4. arr[3]

Q2.
Which value changes?

1. 10
2. 20
3. 30
4. 40

Solution — Modify Through the Pointer!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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`p = arr + 1`

means:

$$\boxed{p \rightarrow arr[1]}$$

Therefore:

`*p = 20`

and:

`*p = *p + 5`

becomes:

$$20 + 5 = 25$$

So the array becomes:

$$\boxed{\begin{array}{cccc} 10 & 25 & 30 & 40 \end{array}}$$

Output: 10 25 30 40

POINTER Q #2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Complete Output!

```
int arr[] = {10,20,30,40};

int *p = arr;

printf("%d ", *p++);

printf("%d ", (*p)++);

printf("%d ", *p);

printf("%d", *(++p));
```

This question contains:

- *
- ++
- p++
- ++p
- (*p)++
- Arrays

Trace the pointer!

Solution!!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Initial state:

10	20	30	40
arr[0]	arr[1]	arr[2]	arr[3]

$p \rightarrow$ arr[0]

Statement 1:

*p++

means: *(p++)

Print 10 $p \rightarrow arr[1]$

Statement 2:

(*p)++

$20 \rightarrow 21$

Statement 3:

*p = 21

LEVEL 4

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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THREE LEVELS OF INDIRECTION!

$$r \longrightarrow q \longrightarrow p \longrightarrow x$$

```
r :  int ***
q :  int **
p :  int *
x :  int
```

Triple Pointer Trap #1

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int x = 25;

int *p = &x;
int **q = &p;
int ***r = &q;

printf("%d", ***r);
```

What is `***r`?

1. Address of x
2. Address of p
3. Address of q
4. Value of x

Solution — Three Dereferences!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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$$r \rightarrow q \rightarrow p \rightarrow x$$

First:

```
*r = q
```

Second:

```
**r = p
```

Third:

```
***r = x
```

Therefore:

```
***r = 25
```

Output: 25

Triple Pointer Trap #2

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int x = 10;

int *p = &x;
int **q = &p;
int ***r = &q;

***r = 99;

printf("%d %d", x, *p);
```

Question:

What does:

```
***r = 99
```

actually modify?

1. r
2. q
3. p
4. x

Solution — ***r Reaches the Actual Variable!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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$$r \rightarrow q \rightarrow p \rightarrow x$$

Therefore:

```
***r = x
```

So:

```
***r = 99
```

is effectively:

```
x = 99
```

Since `p` still points to `x`:

```
*p = 99
```

Output: 99 99

Triple Pointer Trap #3

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int a = 10;
int b = 20;

int *p = &a;
int **q = &p;
int ***r = &q;

**r = &b;

printf("%d", *p);
```

Initially:

$$p \rightarrow a$$

But then:

```
**r = &b
```

Where does `p` point now?

1. a
2. b
3. q
4. r

Solution — `**r` Is `p`!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Remember:

$$r \rightarrow q \rightarrow p$$

Therefore:

```
**r = p
```

So:

```
**r = &b
```

means:

```
p = &b
```

The pointer has been redirected:

$$p \rightarrow b$$

Therefore:

```
*p = 20
```

Output: 20

Triple Pointer #4

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int a = 10;
int b = 20;

int *p = &a;
int *s = &b;

int **q = &p;
int ***r = &q;

*r = &s;

printf("%d", **r);
```

Initially:

$$q \rightarrow p$$

Then:

`*r = &s`

What does q point to now?

1. p
2. s
3. a
4. b

Solution — We Changed q!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Initially:

$$r \to q \to p \to a$$

and:

$$s \to b$$

Now:

```
*r = &s
```

But:

```
*r = q
```

Therefore:

```
q = &s
```

New structure: $r \to q \to s \to b$

Thus:

```
**r = 20
```

Output: 20

Triple Pointer #5

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Predict the Output!

```
int arr[] = {10,20,30};

int *p = arr;
int **q = &p;
int ***r = &q;
printf("%d ", ***r);
(*p)++;
printf("%d ", **q);
(**q)++;
printf("%d", ***r);
```

You need to track:

$$r \rightarrow q \rightarrow p$$

and then:

```
***r
```

Also:

```
(*p)++
```

and:

```
(**q)++
```

Solution— Trace Every Dereference!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Initially: $p \rightarrow arr[0]$

Therefore: `***r = 10`

Then: `(*p)++`

changes: $10 \rightarrow 11$

Now: `**q`

also reaches the same value: `**q = 11`

Then: `(**q)++`

changes: $11 \rightarrow 12$

Therefore: `***r = 12`

$$\boxed{10 \quad 11 \quad 12}$$

Entry Question!

॥ सा विद्या या विमुक्तये ॥
भारतीय प्रौद्योगिकी संस्थान धारवाड
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Coding Challenge

You are given an array of integers.
Write a C program that finds the SECOND LARGEST DISTINCT element in the array.
Example:

$$\texttt{arr} = \{10,\ 5,\ 20,\ 20,\ 8,\ 15,\ 20\}$$

$$\boxed{\text{Answer} = 15}$$

Your program must:

- NOT sort the array
- Use only **one array**
- Work even when the largest value appears multiple times
- Handle the case when a second distinct element does not exist

Can you solve it in ONE traversal?

Entry QUESTION!



Coding Challenge

Given an array of n integers and an integer k,
rotate the array to the RIGHT by k positions. Example:

$$\texttt{arr} = \{1,2,3,4,5,6,7\}$$

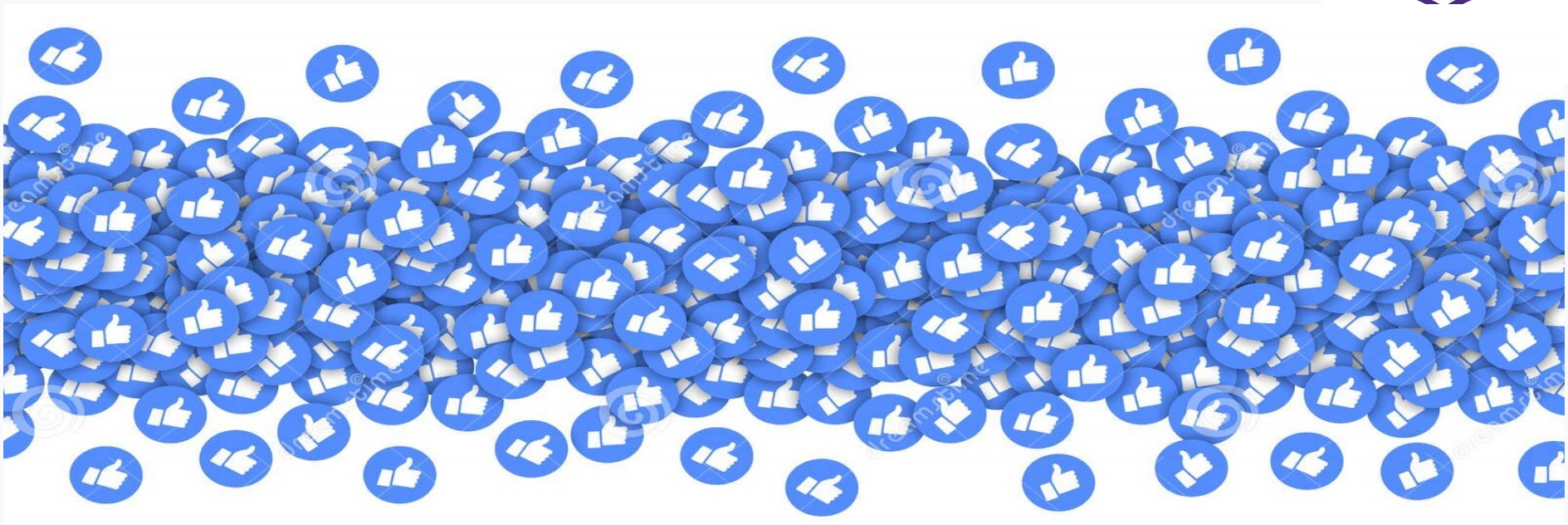
$$k = 3$$

Your program should produce:

$$\boxed{\{5,6,7,1,2,3,4\}}$$

Rules:

- Do **NOT** use another array.
- Do **NOT** use a sorting function.
- k may be larger than n.
- Modify the original array.



Thank you for your kind attention!!